

Assignment No: 6

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Title

Integrate Jenkins with a Git Repository and Automate Source Code Retrieval from GitHub

Aim

To configure Jenkins with a GitHub repository and automate the process of pulling source code from the repository for Continuous Integration.

Theory

Objectives

- Understand the role of Jenkins in CI/CD.
- Install and configure the Git plugin in Jenkins.
- Connect Jenkins with a GitHub repository.
- Create a Jenkins job for automatic source code checkout.
- Execute a simple build through Jenkins.

Continuous Integration and Jenkins

Continuous Integration (CI) is the practice of frequently merging code changes into a shared repository, where every change is automatically built and tested. Jenkins is an open-source automation server written in Java that implements CI and Continuous Delivery (CD) pipelines. Its functionality is extended through plugins – the Git and GitHub plugins let Jenkins clone repositories, and the Pipeline plugin allows the build process to be written as code in a Jenkinsfile.

Developer → GitHub Repository → Jenkins → Build / Test → Build Result

- **Job / Project:** a configured task that Jenkins runs, e.g. a Freestyle project or a Pipeline.
- **Source Code Management (SCM):** where Jenkins gets the code from – here Git with the GitHub repository URL and branch */main.
- **Workspace:** the directory on the Jenkins agent where the repository is cloned and the build runs.
- **Build step:** a command executed after checkout, such as compiling code or running tests.
- **Console Output:** the complete log of a build, including checkout and build results.

- **Jenkinsfile:** a declarative pipeline with stages such as Checkout, Build and Test.

Prerequisites

- Jenkins installed and running (run here as a Docker container, image jenkins/jenkins:lts-jdk17).
- Git and Java/JDK (included in the Jenkins image).
- GitHub account and repository: <https://github.com/tanushkapatil/DevOps>
- Sample application – the Python app from Assignment 5 in the same repository.

Code / Implementation

Step 1: GitHub repository

The public repository <https://github.com/tanushkapatil/DevOps> contains the sample application (Assignment5_MultiContainer/web/app.py), a README.md and the other assignment folders. Because the repository is public, no credentials are required.

Step 2: Install Jenkins with the Git, GitHub and Pipeline plugins

Jenkins was run as a Docker container. The plugins were installed with jenkins-plugin-cli (equivalent to Manage Jenkins → Plugins → Available Plugins → Install) and Jenkins is bound to 127.0.0.1 so it is reachable only from this machine.

Dockerfile – Jenkins with required plugins

```
FROM jenkins/jenkins:lts-jdk17

USER root
# Git and Python are needed by the build steps
RUN apt-get update && apt-get install -y --no-install-recommends python3 git && rm -rf /var/lib/apt/lists/*
USER jenkins

# Required plugins: Git, GitHub, Pipeline (+ stage view)
RUN jenkins-plugin-cli --plugins git github workflow-aggregator pipeline-stage-view

# Skip the setup wizard for this local lab instance
ENV JAVA_OPTS="-Djenkins.install.runSetupWizard=false"
COPY init.groovy.d/ /usr/share/jenkins/ref/init.groovy.d/
```

compose.yaml

```
services:
  jenkins:
    build: .
    container_name: jenkins
    ports:
      - "127.0.0.1:8090:8080" # reachable from this machine only
    volumes:
      - jenkins_home:/var/jenkins_home

volumes:
  jenkins_home:
```

```
docker compose build
docker compose up -d
# Jenkins UI: http://127.0.0.1:8090
```

Step 3: Create a Jenkins job

- Open the Jenkins Dashboard → New Item.
- Enter the job name GitHub-Jenkins-Demo.
- Select Freestyle project and click OK.

Step 4: Configure Source Code Management

- Source Code Management → Git.
- Repository URL: <https://github.com/tanushkapatil/DevOps.git>
- Credentials: none (public repository).
- Branch Specifier: */main

Step 5: Configure the build step

Build Steps → Add build step → Execute shell:

```
echo "=== Workspace contents ==="
ls -la
echo "=== Latest commits ==="
git log --oneline -3
echo "=== Build: compile Python app ==="
python3 --version
python3 -m py_compile Assignment5_MultiContainer/web/app.py && echo "app.py compiled successfully"
```

Step 6: Build the job

Click Build Now, then open the build and select Console Output to verify the checkout and build.

Optional: Jenkins Pipeline

A second job, GitHub-Jenkins-Pipeline (type Pipeline), uses the following Jenkinsfile:

Jenkinsfile

```
pipeline {
    agent any

    stages {
        stage('Checkout') {
            steps {
                git branch: 'main',
                    url: 'https://github.com/tanushkapatil/DevOps.git'
            }
        }

        stage('Build') {
            steps {
                echo 'Building application...'
            }
        }
    }
}
```

```

        sh 'python3 -m py_compile Assignment5_MultiContainer/web/app.py'
        sh 'ls -la'
    }
}

stage('Test') {
    steps {
        echo 'Running tests...'
        sh 'test -f README.md && echo "README found"'
        sh 'git log --oneline -3'
    }
}
}
}

```

Output

1. Jenkins container running, with Git, Java and Python available:

```

Windows PowerShell
PS C:\Users\yogir\OneDrive\Desktop\Devops\Assignment6_Jenkins> docker compose build
...
#12 DONE 0.4s
Image assignment6_jenkins-jenkins Built

PS C:\Users\yogir\OneDrive\Desktop\Devops\Assignment6_Jenkins> docker compose up -d
Container jenkins Starting
Container jenkins Started

PS C:\Users\yogir\OneDrive\Desktop\Devops\Assignment6_Jenkins> docker compose ps
NAME                IMAGE                                STATUS      PORTS
jenkins             assignment6_jenkins-jenkins         Up 5 minutes  127.0.0.1:8090->8080/tcp

PS C:\Users\yogir\OneDrive\Desktop\Devops\Assignment6_Jenkins> docker exec jenkins git --version
git version 2.47.3

PS C:\Users\yogir\OneDrive\Desktop\Devops\Assignment6_Jenkins> docker exec jenkins java -version 2>&
1 | Select-Object -First 1
openjdk version "17.0.18" 2026-01-20

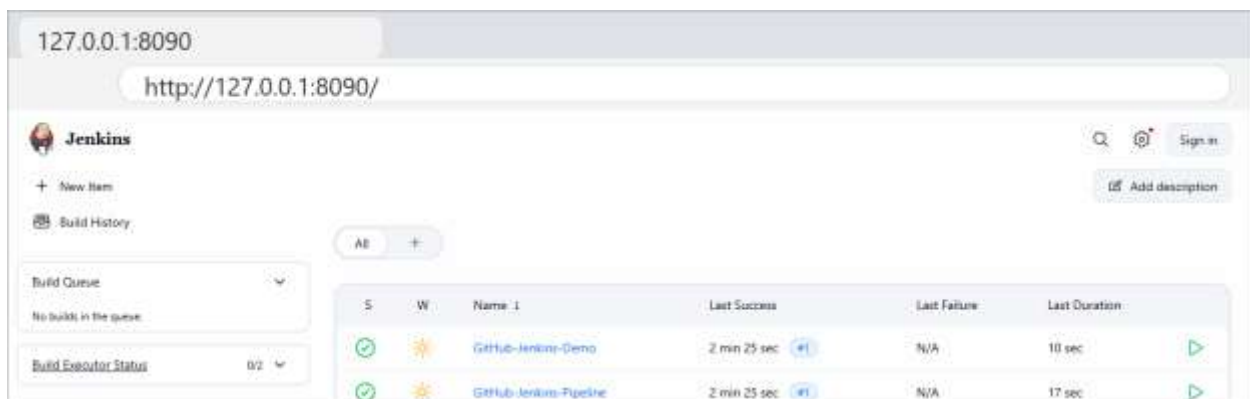
PS C:\Users\yogir\OneDrive\Desktop\Devops\Assignment6_Jenkins> docker exec jenkins python3 --version
Python 3.13.5

```

2. Git, GitHub and Pipeline plugins installed and active:

```
Windows PowerShell
PS C:\Users\yogir\OneDrive\Desktop\Devops\Assignment6_Jenkins> python list-plugins.py
PLUGIN          NAME                VERSION             ACTIVE
git             Git plugin          5.10.1              True
git-client      Jenkins Git client  6.6.1               True
github          GitHub plugin       1.47.0              True
github-api      GitHub API Plugin   1.330-492.v3941a_032db True
workflow-aggregator Pipeline           608.v67378e9d3db_1 True
workflow-job     Pipeline: Job       1571.1580.v18e46842c12 True
pipeline-stage-view Pipeline: Stage View Plugin 2.41                True
```

3. Jenkins dashboard showing both jobs with successful builds:



4. Source Code Management configuration of GitHub-Jenkins-Demo (Git, repository URL, branch */main):

The screenshot shows the Jenkins web interface for configuring a job. The browser address bar displays 'http://127.0.0.1:8090/job/GitHub-Jenkins-Demo/configure'. The page title is 'Source Code Management'. Below the title, a subtitle reads 'Connect and manage your code repository to automatically pull the latest code for your builds.' There are two radio buttons: 'None' and 'Git', with 'Git' selected. Under the 'Git' section, there is a 'Repository' field with a dropdown arrow. Below it is a 'Repository URL' text box containing 'https://github.com/tenushkapati/DevOps.git'. Below that is a 'Credentials' dropdown menu showing '- none -' and an 'Add' button. An 'Advanced' dropdown menu is also present. Below the repository section is a '+ Add Repository' button. The 'Branches to build' section has a 'Branch Specifier (blank for \'any\')' text box containing '*/main'. Below this is a '+ Add Branch' button. The 'Repository browser' section has a dropdown menu showing '(Auto)'. At the bottom, the 'Additional Behaviours' section has a '+ Add' button.

127.0.0.1:8090

http://127.0.0.1:8090/job/GitHub-Jenkins-Demo/configure

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ Git

Repository

Repository URL

https://github.com/tenushkapati/DevOps.git

Credentials

- none -

Add

Advanced

+ Add Repository

Branches to build

Branch Specifier (blank for 'any')

*/main

+ Add Branch

Repository browser

(Auto)

Additional Behaviours

+ Add

5. Console Output of build #1 – repository cloned from GitHub, latest commit checked out, build step executed, Finished: SUCCESS:

The screenshot shows the Jenkins web interface for a build named 'GitHub-Jenkins-Demo/1'. The address bar shows 'http://127.0.0.1:8090/job/GitHub-Jenkins-Demo/1/console'. The left sidebar contains links for 'Status', 'Changes', 'Console Output', 'Edit Build Information', 'Delete build WT', and 'Git Build Data'. The 'Console Output' tab is active, displaying the following text:

```
Started by user unknown or anonymous
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/GitHub-Jenkins-Demo
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/tanushkapatil/DevOps.git
> git init /var/jenkins_home/workspace/GitHub-Jenkins-Demo # timeout=10
fetching upstream changes from https://github.com/tanushkapatil/DevOps.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/tanushkapatil/DevOps.git +refs/heads/*:refs/remotes/origin/* #
timeout=10
> git config remote.origin.url https://github.com/tanushkapatil/DevOps.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rebase refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 11070562981a3d8a7adcb25082f6015be70554 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 11070562981a3d8a7adcb25082f6015be70554 # timeout=10
Commit message: "Assignment 5: larger, more readable screenshots"
First time build. Skipping changelog.
[GitHub-Jenkins-Demo] $ /bin/sh -xe /tmp/jenkins18521289093929067451.sh
+ echo === Workspace contents ===
=== Workspace contents ===
+ ls -la
total 1008
drwxr-xr-x 5 jenkins jenkins 4096 Sep 26 17:54 .
drwxr-xr-x 5 jenkins jenkins 4096 Sep 26 17:54 ..
drwxr-xr-x 8 jenkins jenkins 4096 Sep 29 17:54 .git
-rw-r--r-- 1 jenkins jenkins 104 Sep 29 17:54 .gitignore
-rw-r--r-- 1 jenkins jenkins 61047 Sep 29 17:54 Assignment1Template.pdf
-rw-r--r-- 1 jenkins jenkins 81072 Sep 29 17:54 Assignment2_Cloud_Computing_AWS.docx
-rw-r--r-- 1 jenkins jenkins 148099 Sep 29 17:54 Assignment2_Cloud_Computing_AWS.pdf
drwxr-xr-x 3 jenkins jenkins 4096 Sep 29 17:54 Assignment3_Terraform
-rw-r--r-- 1 jenkins jenkins 479601 Sep 29 17:54 Assignment3_Terraform.docx
-rw-r--r-- 1 jenkins jenkins 120171 Sep 29 17:54 Assignment3_Terraform.pdf
drwxr-xr-x 3 jenkins jenkins 4096 Sep 29 17:54 Assignment5_MultiContainer
-rw-r--r-- 1 jenkins jenkins 449026 Sep 29 17:54 Assignment5_MultiContainer.docx
-rw-r--r-- 1 jenkins jenkins 351507 Sep 29 17:54 Assignment5_MultiContainer.pdf
-rw-r--r-- 1 jenkins jenkins 1000 Sep 29 17:54 README.md
+ echo === latest commits ===
=== latest commits ===
+ git log --oneline -3
1107056 Assignment 5: larger, more readable screenshots
a0c5c61 Assignment 2: add figure placeholders for EC2 screenshots
10f6b20 Add Github repository links to reports
+ echo === Build: compile Python app ===
=== Build: compile Python app ===
+ python3 --version
Python 3.11.5
+ python3 -m py_compile Assignment5_MultiContainer/web/app.py
+ echo app.py compiled successfully
App.py compiled successfully
Finished: SUCCESS
```

6. Workspace containing the latest source code pulled from GitHub:

The screenshot shows the Jenkins web interface for a workspace named 'GitHub-Jenkins-Demo'. The browser address bar displays 'http://127.0.0.1:8090/job/GitHub-Jenkins-Demo/ws/'. The Jenkins logo and 'Sign in' button are visible in the top right. A left sidebar contains navigation links: Status, Changes, Workspace (selected), Wipe Out Current Workspace, Build Now, Configure, Delete Project, and Rename. Below these is a 'Builds' section with a filter input and a status bar showing 'Today' with a green circle and '11:54 PM'. The main area is titled 'Workspace of GitHub-Jenkins-Demo on Built-In Node' and features a search bar. Below the search bar is a list of files and folders:

File/Folder	Size	Modified
git		
Assignment1_Terraform		
Assignment5_MultiContainer		
gitignore	104 B	Sep 29, 2025, 5:54:20 PM
Assignment2_Cloud_Computing_AWS.docx	60.52 KB	Sep 29, 2025, 5:54:20 PM
Assignment2_Cloud_Computing_AWS.pdf	145.11 KB	Sep 29, 2025, 5:54:20 PM
Assignment1_Terraform.docx	488.36 KB	Sep 29, 2025, 5:54:20 PM
Assignment3_Terraform.pdf	521.46 KB	Sep 29, 2025, 5:54:20 PM
Assignment5_MultiContainer.docx	438.50 KB	Sep 29, 2025, 5:54:20 PM
Assignment5_MultiContainer.pdf	343.35 KB	Sep 29, 2025, 5:54:20 PM
Assignment_Template.pdf	59.62 KB	Sep 29, 2025, 5:54:20 PM
README.md	1.00 KB	Sep 29, 2025, 5:54:20 PM

At the bottom of the file list, there is a link: (all files in zip).

7. Pipeline job – Stage View showing the Checkout, Build and Test stages completed successfully:

127.0.0.1:8090

http://127.0.0.1:8090/job/GitHub-Jenkins-Pipeline/

Jenkins GitHub-Jenkins-Pipeline

Status

Changes

Build Now

Configure

Delete Pipeline

Full Stage View

Rename

Pipeline Syntax

Builds

Filter

1 1:54 PM

GitHub-Jenkins-Pipeline

Declarative pipeline: Checkout, Build and Test stages

Stage View

Average stage times: (Full run time: ~17s)

	Checkout	Build	Test
Geo 20 (2024)	3s	7s	984ms

Permalinks

- Last build (#1), 2 min 58 sec ago
- Last stable build (#1), 2 min 58 sec ago
- Last successful build (#1), 2 min 58 sec ago
- Last completed build (#1), 2 min 58 sec ago

Windows PowerShell

```
PS C:\Users\yogir\OneDrive\Desktop\Devops\Assignment6_Jenkins> curl.exe -s http://127.0.0.1:8090/job/GitHub-Jenkins-Pipeline/1/consoleText (key lines)
Running on Jenkins in /var/jenkins_home/workspace/GitHub-Jenkins-Pipeline
[Pipeline] stage
[Pipeline] { (Checkout)
Cloning repository https://github.com/tanushkapatil/DevOps.git
Checking out Revision 11070562b01e3a0a7adccb25082fd015b6e7b654 (refs/remotes/origin/main)
Commit message: "Assignment 5: larger, more readable screenshots"
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building application...
+ python3 -m py_compile Assignment5_MultiContainer/web/app.py
+ ls -la
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Running tests...
+ test -f README.md
+ echo README found
README found
+ git log --oneline -3
Finished: SUCCESS
```

Conclusion

In this assignment, I learned about integrating Jenkins with GitHub for Continuous Integration. I ran Jenkins with the Git, GitHub and Pipeline plugins, created a Freestyle job that checks out the main branch of my GitHub repository and runs a build step, and verified in the console output and workspace that the latest source code was retrieved and built successfully. I also created a declarative pipeline with Checkout, Build and Test stages, which shows how every code change can flow automatically through a CI/CD workflow.

References

1. Jenkins User Documentation – <https://www.jenkins.io/doc/>
2. Jenkins Git Plugin – <https://plugins.jenkins.io/git/>
3. Pipeline Syntax – <https://www.jenkins.io/doc/book/pipeline/syntax/>
4. Jenkins Docker Image – <https://github.com/jenkinsci/docker/blob/master/README.md>
5. GitHub Repository used – <https://github.com/tanushkapatil/DevOps>

Submission Links

GitHub Repository Link:

https://github.com/tanushkapatil/DevOps/tree/main/Assignment6_Jenkins

Hosted / Output Link: <http://127.0.0.1:8090> (local Jenkins instance)